LUC MCCUTCHEON

RESEARCH ENGINEER · VLA POST-TRAINING

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Research Engineer and PhD Candidate (completing Summer 2026) specialising in RL for distributed VLA post-training. Co-Founder/CTO of GeoSnap. Expert in JAX/PyTorch. Published at ICRA, NeurIPS, and IROS.

RESEARCH SCIENTIST (FIXED-TERM)

Cambridge Consultants — Oct 2025 – Mar 2026

- Developed reinforcement learning pipelines for the Unitree G1 humanoid robot
- Fine-tuned Vision-Language-Action models (GR00T) for manipulation tasks
- Built simulation environments in MuJoCo and NVIDIA Isaac Sim
- Designed and executed sim-to-real transfer experiments

LEAD RESEARCH SCIENTIST (PART-TIME)

Agile Loop — Oct 2023 – Apr 2025

- Led a team of 10 researchers across multiple agentic AI projects
- Developed agentic pipelines combining RL and VLM LoRA fine-tuning
- Presented technical results and product demos to Google Cloud stakeholders

RESEARCH SCIENTIST (PART-TIME)

Agile Loop — Jun 2023 – Oct 2023

- Built RL environments for software testing and evaluation
- Developed computer vision systems for product applications
- Designed hybrid edge/cloud model routing architecture for Lenovo

ASSISTANT RESEARCH VOLUNTEER

Connected & Autonomous Vehicle Lab — Jun 2022 – Aug 2022

- Implemented Noisy Neural Networks and Deep Q-Learning for autonomous driving
- Developed LSTM-based prediction models for vehicle trajectory forecasting

SOFTWARE ENGINEER INTERN

QinetiQ — Jul 2018–Sep 2018 & Jul 2019–Sep 2019

- Developed voice cryptography systems in C++ and Python
- Conducted red team security assessments and penetration testing

PHD REINFORCEMENT LEARNING (SPONSORED)

University of Surrey — 2021 – Present

Research focus: world models, Neural Lyapunov functions for stability, time-delay control systems, and large-scale JAX implementations. Supervised research with publications at top robotics and ML venues.

BSC COMPUTER SCIENCE (HONS) — FIRST CLASS

University of Surrey — 2018 – 2021

CyberFirst Scholarship recipient. Strong foundation in algorithms, systems, and applied machine learning.

Frameworks & Libraries: JAX, PyTorch, Gymnasium, vLLM, NumPy

Languages: Python, C++, Rust, JavaScript

RL & ML: PPO, GRPO, LoRA, VLA, SAC, RAINBOW, ResNet

Infrastructure & Tools: Unitree G1, SLURM, GCP, AWS, Docker

Languages (Human): French (C1)

First Author

- **Efficient Neural Lyapunov Function Approximation with Off-Policy Reinforcement Learning** — ICRA 2025
- **Adaptive PD Control using Deep Reinforcement Learning** — IROS 2023
- **Learning Continually at Peak Performance** — Under Review

Co-Author

- **Meta-World+: Towards a Scalable Multi-Task Reinforcement Learning Benchmark** — ICML 2025 Workshop (Spotlight) & NeurIPS 2025
- **Exploring In-Context Ensemble Strategies for Reinforcement Learning** — NeurIPS 2024 Workshop
- **Prediction-Based Decision Making for Autonomous Vehicles** — ITSC 2022

Reviewing

ICLR 2026 | NeurIPS 2025 | TNNLS 2025 | IROS 2025

- Honourable Mention — Grokathon 2026
- Bronze Medal — Mathematics & Informatics Olympiad
- Grace Hopper Award — University of Surrey
- Foundership Award
- NVIDIA Deep Learning Certificate
- Coursera Reinforcement Learning Specialisation
- Extended Project Qualification (EPQ)
- **Google Agents Workshop** — Invited talk on agentic AI and reinforcement learning
- **University of Surrey** — Research talks on Neural Lyapunov functions and continual RL
- **Cambridge Consultants** — Technical talks on humanoid robotics and sim-to-real transfer